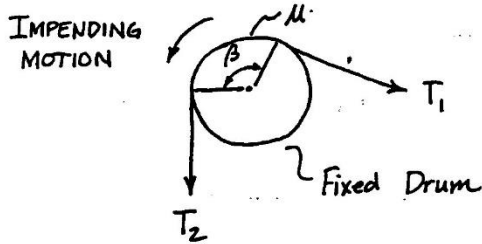


FRICTION ON FLAT BELTS (As opposed to V-belts)



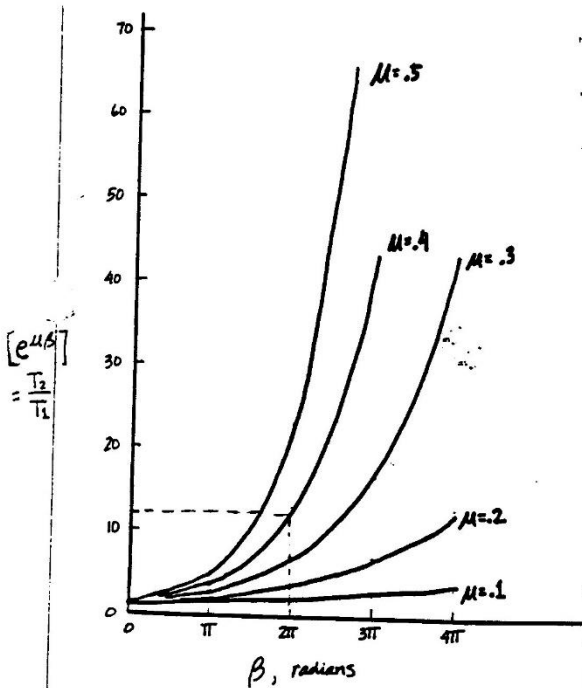
$T_2 > T_1$ due to friction

$$T_2 = T_1 e^{\mu\beta}$$

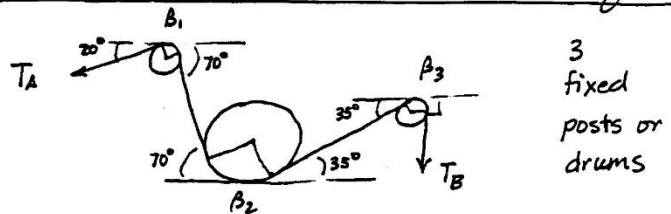
$\mu \equiv$ Coeff. of friction between belt and fixed drum.

$\beta \equiv$ angle of belt-to-surface contact in units of radians.

Properties of this exponential function:



NOTICE that the radius of the fixed drum is not a variable in this eqn.



$$\beta_1 = \frac{20^\circ + 70^\circ}{180} \pi = \frac{\pi}{2} = 1.57 \text{ rad}$$

$$\beta_2 = \frac{70^\circ + 35^\circ}{180} \pi = 1.83 \text{ rad}$$

$$\beta_3 = \frac{90^\circ + 35^\circ}{180} \pi = 2.18 \text{ rad}$$

3
fixed
posts or
drums

If motion is impending from B to A, then $T_A > T_B$,

$$T_A = e^{\sum(\mu\beta)} T_B$$

If μ is the same for each fixed drum:

$$T_A = e^{\mu(\sum\beta)} T_B$$

Example: $\mu = 0.4$

	$\beta = 2\pi$ (1 wrap)	$\beta = 4\pi$ (2 wraps)
$\frac{T_2}{T_1}$	= 12.3	152.4
	$\approx 123 \text{ lb}$	$\approx 1500 \text{ lb}$

Example Applications: Using powered capstans to move very massive objects

